



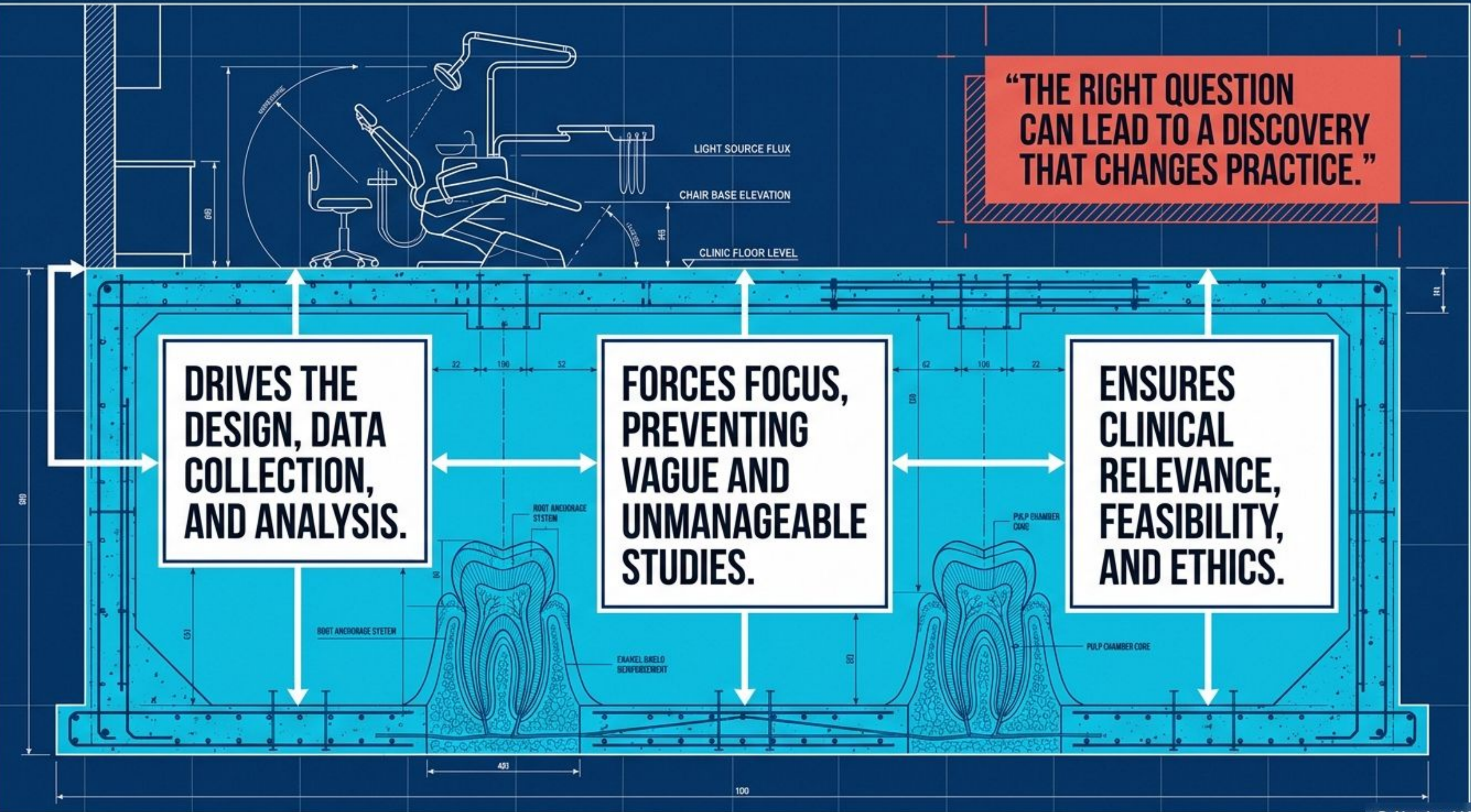
FORMULATING QUESTIONS, HYPOTHESES, AND OBJECTIVES

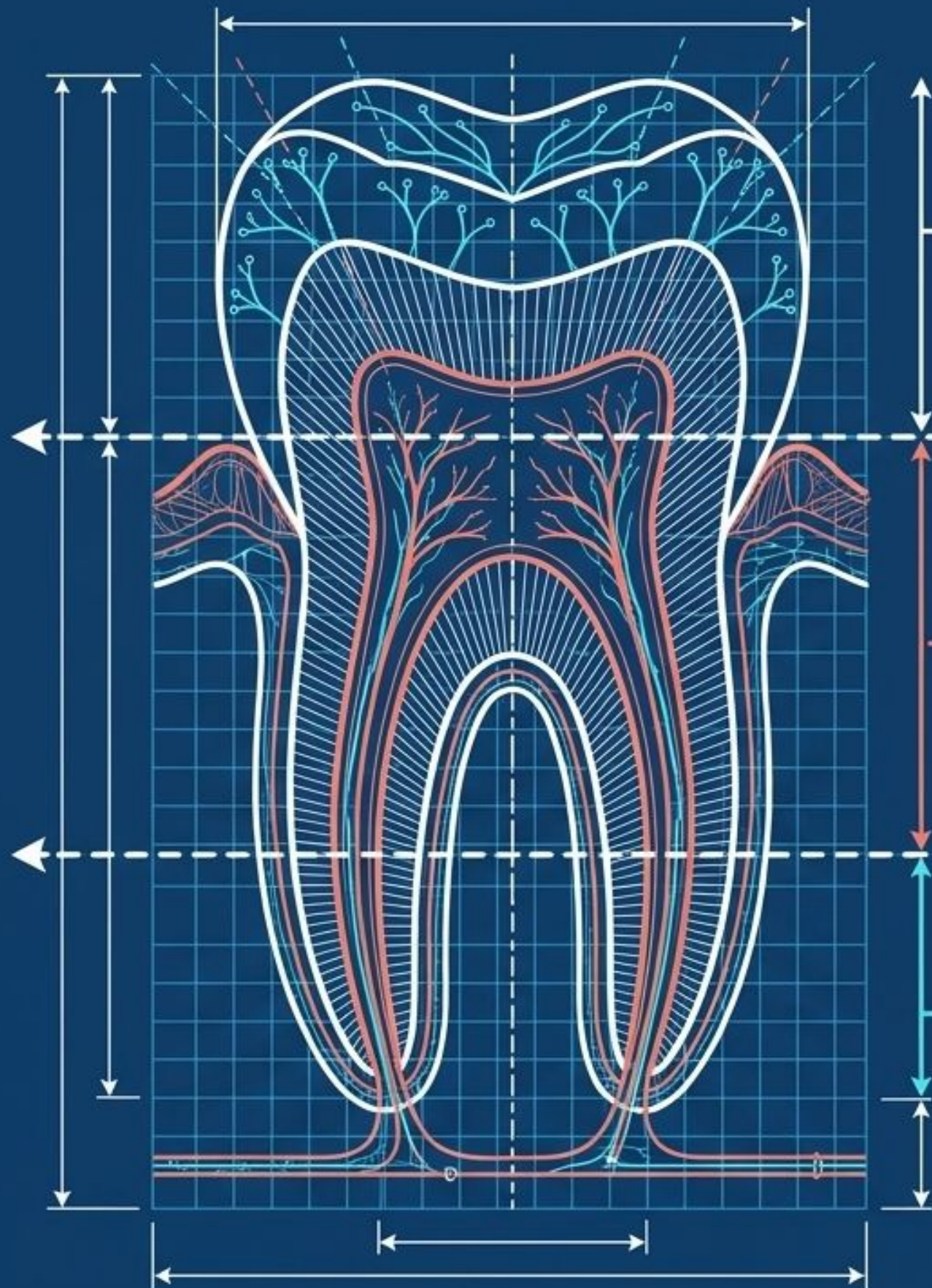
**“THE RIGHT QUESTION
CAN LEAD TO A DISCOVERY
THAT CHANGES PRACTICE.”**

**DRIVES THE
DESIGN, DATA
COLLECTION,
AND ANALYSIS.**

**FORCES FOCUS,
PREVENTING
VAGUE AND
UNMANAGEABLE
STUDIES.**

**ENSURES
CLINICAL
RELEVANCE,
FEASIBILITY,
AND ETHICS.**





THE CROWN: THE OBJECTIVES

The visible, actionable outputs.
Measurable steps taken to test the hypothesis.

THE TRUNK: THE HYPOTHESIS

The core structure. The testable prediction bridging the question to the action.

THE ROOTS: THE RESEARCH QUESTION

The deep, hidden foundation.
Clear, concise, and arguable.

**GAPS IN CURRENT
CLINICAL PRACTICE.**

**INTERESTING ANOMALIES
ENCOUNTERED
IN THE CLINIC.**

**DEBATES IN CURRENT
JOURNALS AND
CASE STUDIES.**

**DISCUSSIONS WITH
SENIOR CLINICIANS
AND TEACHERS.**

CORE DIRECTIVE: CHOOSE A TARGET THAT IS NARROW AND PERFECTLY MANAGEABLE.



DESCRIPTIVE

What is
happening?



COMPARATIVE

Is there a
difference?



ANALYTICAL

Why is it
happening?



EXPLORATORY

What are
the factors?

RAW CLINICAL IDEA

F - FEASIBLE:

I - INTERESTING:

N - NOVEL:

E - ETHICAL:

R - RELEVANT:

Can you actually do it?

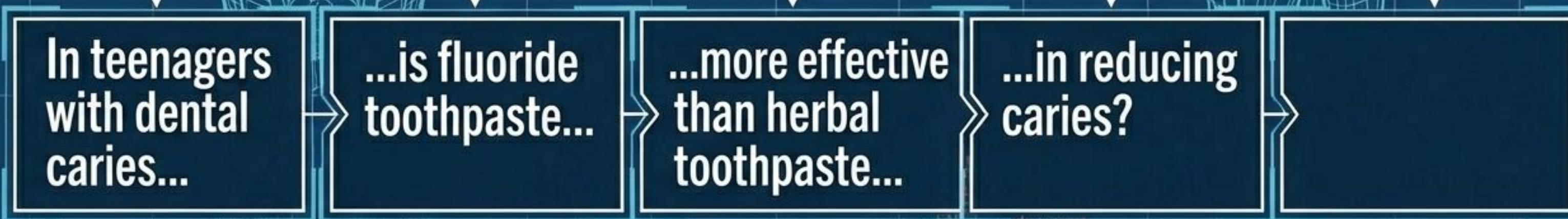
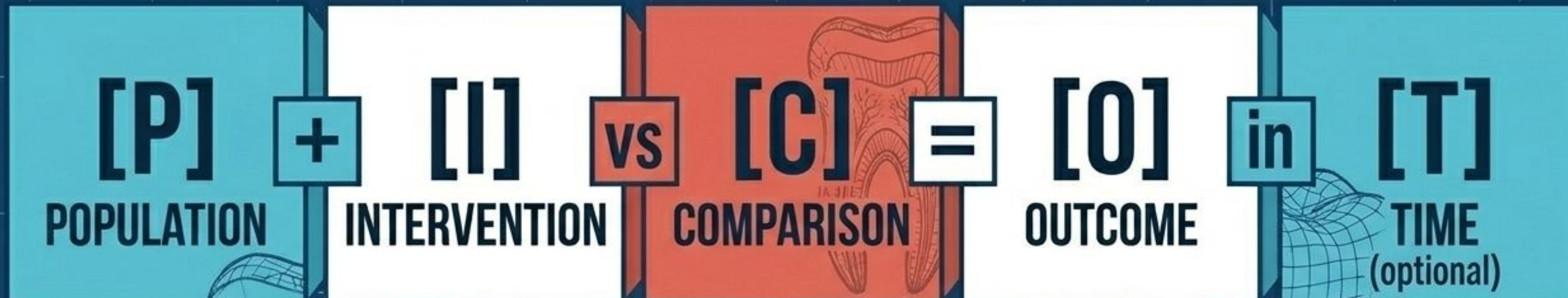
Does it engage you and peers?

Does it add new knowledge?

Does it protect participants?

Does it impact the field?

THE RESEARCH QUESTION





Is dental health important?

Diagnosis: Too broad, unanswerable, vague.



Does regular scaling reduce gingivitis in college students aged 18–22?



Diagnosis: Specific population, clear intervention, measurable outcome.



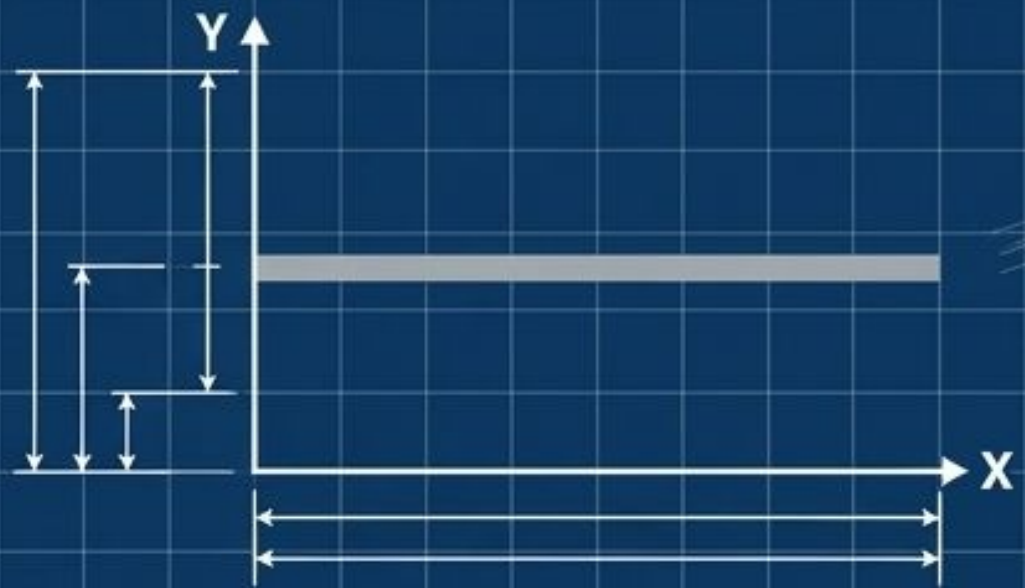
Core Definition: An educated guess or proposed explanation for a phenomenon.

1. Based on prior knowledge, observation, or literature.

2. Predicts a direct relationship between variables.

3. Must be strictly testable and falsifiable.

Working Example: Using fluoride toothpaste reduces dental caries in children.



NULL HYPOTHESIS (H_0)

No difference or relationship exists.

Applied Example: There is no difference in caries incidence between children using fluoride and herbal toothpaste.



ALTERNATIVE HYPOTHESIS (H_1)

A distinct difference or relationship does exist.

Applied Example: Children using fluoride toothpaste have fewer caries than those using herbal toothpaste.

Aim: Broad Overall Goal

To assess the effectiveness of fluoride toothpaste in preventing caries.

1. To compare caries prevalence in users of fluoride vs. non-fluoride toothpaste.

Narrow, Objectives:
Narrow, Actionable Steps

2. To measure caries reduction after 3 months of use.

S - Specific

M - Measurable

A - Achievable

R - Realistic

T - Time-bound

S

M

A

R

T

To measure caries reduction (S, M) using DMFT indices (A, R) after 3 months of use (T).

**TO
COMPARE**

**TO
ASSESS**

**TO
EVALUATE**

**TO
IDENTIFY**

**TO
DETERMINE**

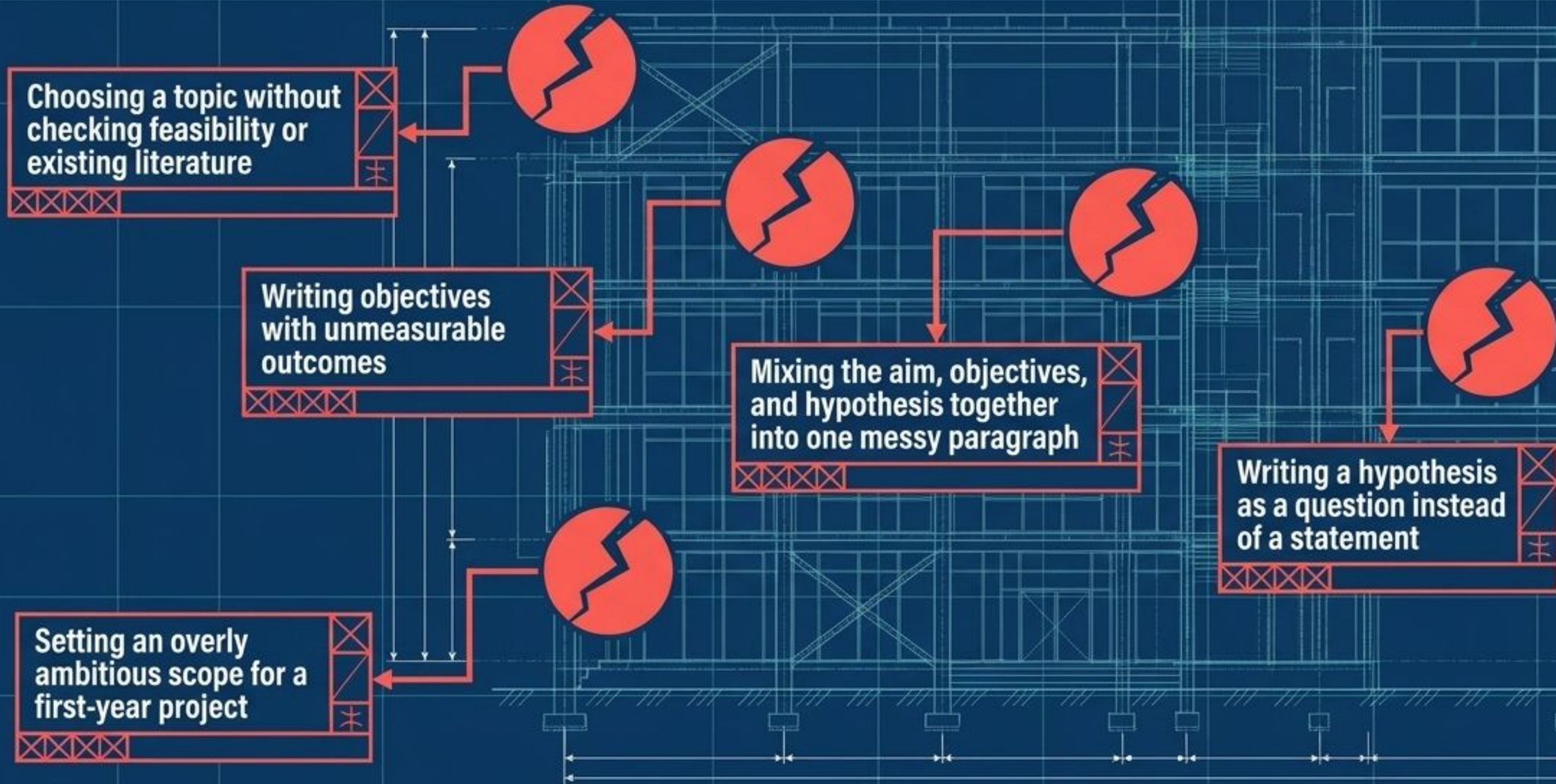
**To evaluate the antimicrobial effect of
chlorhexidine mouthwash on oral biofilm.**

**HYPOTHESIS:
CHLORHEXIDINE
MOUTHWASH
SIGNIFICANTLY
REDUCES ORAL
BIOFILM.**

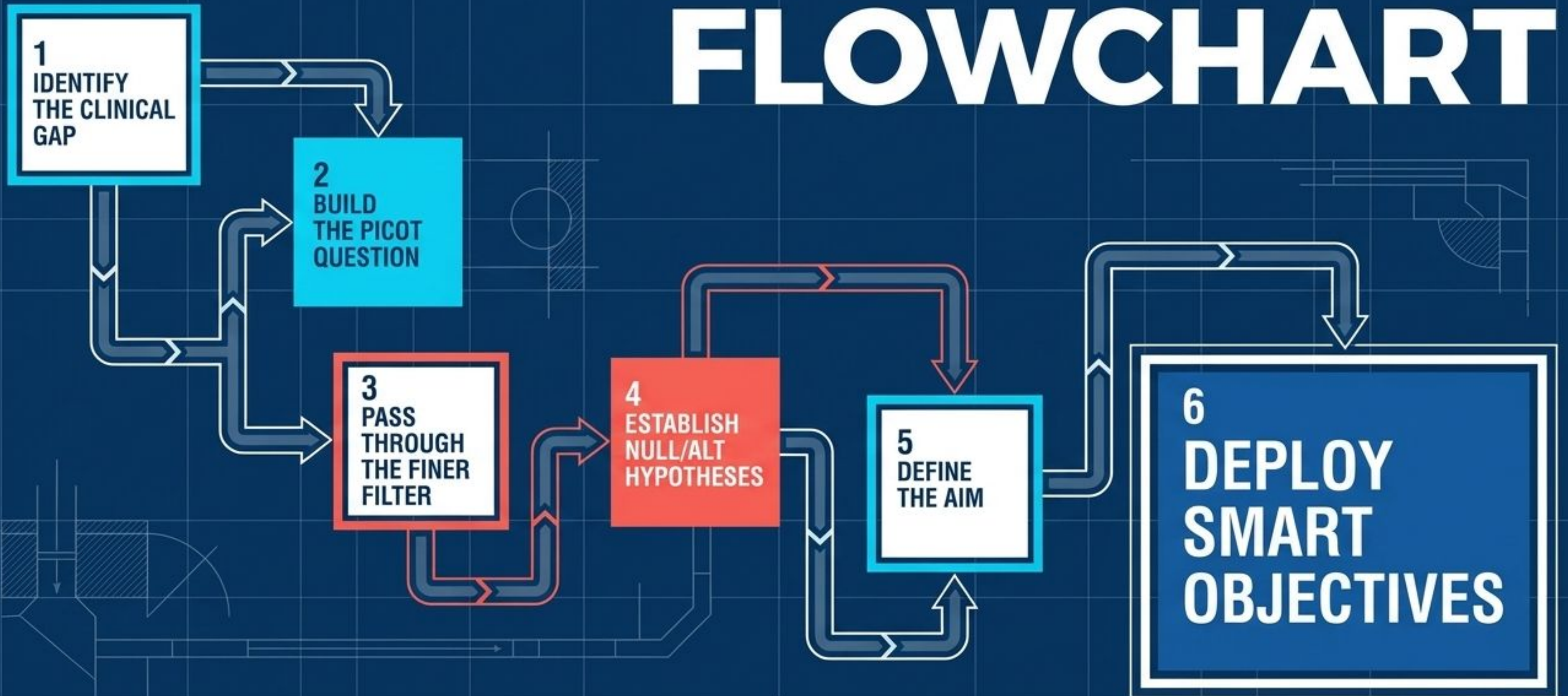
**OBJECTIVE 1:
TO MEASURE
BIOFILM SCORE
BEFORE AND
AFTER USE.**

**OBJECTIVE 2:
TO COMPARE
BIOFILM REDUCTION
WITH AND WITHOUT
MOUTHWASH.**

STRUCTURAL FAILURES



MASTERSYNTHESIS FLOWCHART



Think-Pair-Share: Draft the Blueprint

For your chosen topic, write 1 Hypothesis and 2 SMART Objectives. Discuss, refine, and test against the FINER sieve.

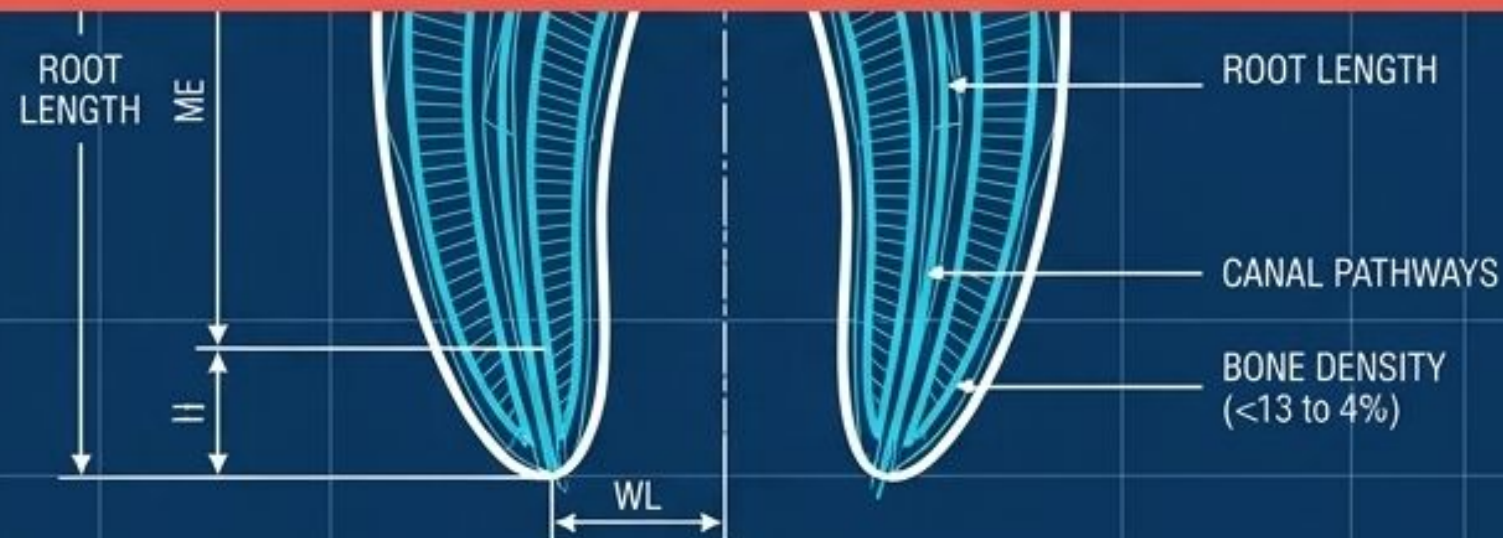
**Target A: Dental
caries in children.**

**Target B: Efficacy
of mouthwashes.**

**Target C: Smoking
and oral lesions.**



**Well-formed questions and clear objectives
are the roots of every good study.**



Keep practicing. / Start small. / Think logically.